Supplementary Material

Supplementary Material for "Few-Shot Specific Emitter Identification: A Signal-Type Mismatched Learning Approach"

I. Sensitivity Analysis of Key Hyperparameters

We report the sensitivity analyses of the key hyperparameters of the proposed method, including the masking ratio ρ, the perturbation coefficient λ, the FDP activation probability p, the aggregation depth L, and the self-distillation strategy. All analyses are conducted on the WiFi dataset under the 20-shot mismatched setting (MS), i.e., the model is pre-trained on ADS-B and fine-tuned on WiFi, and each accuracy value is averaged over the 50 random seeds. The results are summarized in Figs. S1–S5.

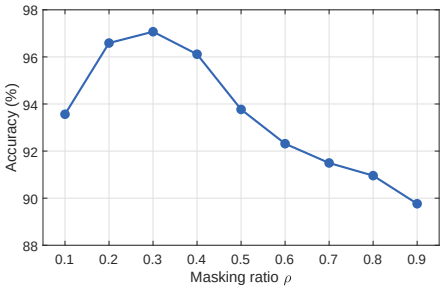


Fig. S1: Sensitivity of masking ratio ρ (WiFi, 20-shot, MS).

A. Masking Ratio ρ

On the WiFi dataset (20-shot, MS), the masking ratio is varied from 0.1 to 0.9 with a step of 0.1. The identification accuracy first increases and then decreases, reaching the maximum at $\rho = 0.3$ (97.07%); the accuracy is 93.56% at $\rho = 0.1$ and drops to 89.76% at $\rho = 0.9$ as the masking becomes too aggressive. A ratio that is too small leaves the pretext task trivial, whereas one that is too large removes so much context that the latent target becomes unrecoverable; the optimum at $\rho = 0.3$ therefore reflects a balance between pretext-task difficulty and retained context. The results are shown in Fig. S1.

B. Perturbation Coefficient λ

On the WiFi dataset (20-shot, MS), the coefficient λ is varied from 0.01 to 0.9, with a step of 0.01 below 0.1 and a step of 0.1 above, and the working point is $\lambda = 0.05$. The accuracy peaks exactly at the working point (97.07%), with 96.13% at $\lambda = 0.04$, and then decreases as λ increases, i.e., as the perturbation weakens, dropping to 87.23% at $\lambda = 0.9$. Because λ weights the instance's own statistics against those of a shuffled instance, a small λ (i.e., strong perturbation) prevents the encoder from relying on auxiliary-data-specific statistics, which is precisely the effect the FDP is designed for. The results are shown in Fig. S2.

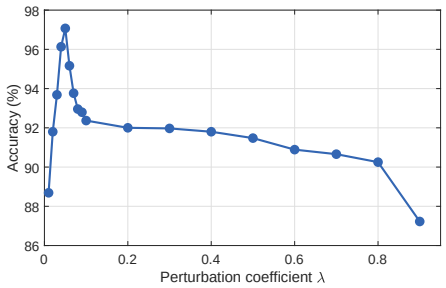


Fig. S2: Sensitivity of perturbation coefficient λ (WiFi, 20-shot, MS).

C. FDP Activation Probability p

On the WiFi dataset (20-shot, MS), the activation probability is varied from 0.1 to 0.9 with a step of 0.1 and is optimal at $p = 0.5$ (97.07%), with the accuracy staying within 92.11–97.07% over the whole range. Activating the perturbation on roughly half of the forward passes is the most favourable setting: a lower p perturbs the auxiliary-specific statistics insufficiently, whereas a higher p discards too much instance-specific information. The results are shown in Fig. S3.

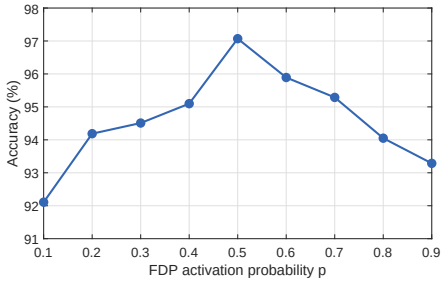


Fig. S3: Sensitivity of FDP activation probability p (WiFi, 20-shot, MS).

D. Aggregation Depth L

On the WiFi dataset (20-shot, MS), the aggregation depth is varied from 1 to 9 layers. The accuracy increases overall with L, from 90.24% at $L = 1$ to 97.07% at $L = 9$; in particular, it reaches 96.91% at $L = 7$, and aggregating all nine layers gives the best result, confirming the benefit of multi-layer aggregation. Most of the gain is obtained once the shallow layers are included ($L = 1 \rightarrow 7$), suggesting that early-layer features carry complementary RFF information that the deepest layer alone does not encode, after which the gains saturate. The results are shown in Fig. S4.

E. Self-Distillation Strategy

On the WiFi dataset (20-shot, MS), the EMA-teacher variant (the proposed design) achieves 97.07±1.47%, outperforming the stop-gradient-only variant (96.23±1.92%) and the variant without distillation (93.82±2.53%). This indicates that the EMA teacher supplies a more stable target than its stop-gradient counterpart, and that the warm-up stage itself, rather than the EMA update alone, accounts for much of the gain. The results are shown in Fig. S5.

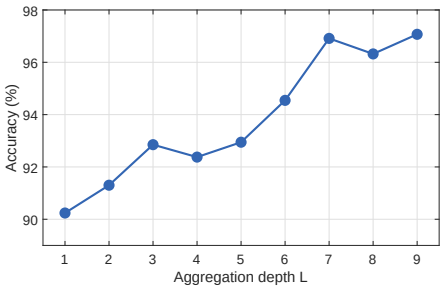


Fig. S4: Sensitivity of aggregation depth L (WiFi, 20-shot, MS).

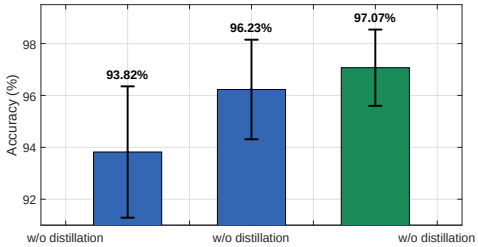


Fig. S5: Self-distillation strategy ablation (WiFi, 20-shot, MS).

II. Reversed Pre-training/Downstream Combinations

To demonstrate the general applicability beyond the ADS-B-based auxiliary pre-training, we additionally conduct experiments with reversed pre-training/downstream combinations, i.e., LoRa $\rightarrow$ WiFi, WiFi $\rightarrow$ LoRa, LoRa $\rightarrow$ ADS-B, and WiFi $\rightarrow$ ADS-B, under the 20-shot setting. The results are summarized in Table S1, which show that the proposed method consistently outperforms AMAE under all reversed combinations (by 1.43–6.47% in absolute accuracy). The absolute accuracy differs across directions because the downstream datasets differ in size (10 categories for ADS-B, 20 for LoRa, and 16 for WiFi) and in signal characteristics, whereas the margin over AMAE remains positive in all four directions.

TABLE S1: Performance under reversed pre-training/downstream combinations (20-shot).

Methods	LoRa$\rightarrow$WiFi	WiFi$\rightarrow$LoRa	LoRa$\rightarrow$ADS-B	WiFi$\rightarrow$ADS-B
Proposed	96.83$\pm$1.77%	90.13$\pm$3.16%	82.41$\pm$2.53%	83.29$\pm$1.94%
AMAE	90.36$\pm$1.58%	85.37$\pm$2.31%	80.98$\pm$2.17%	81.35$\pm$1.66%

III. Robustness under Different SNRs

To evaluate the robustness to practical wireless impairments, the test signals of the WiFi dataset (20-shot, MS) are corrupted by additive white Gaussian noise at SNRs from 0 to 30 dB with a step of 5 dB. As shown in Fig. S6, the accuracy stays above 94% for SNR $\geq$ 20 dB and degrades sharply at lower SNRs, reaching 20.38% at 0 dB. The accuracy thus increases steadily with the SNR and saturates once the noise floor is sufficiently low.

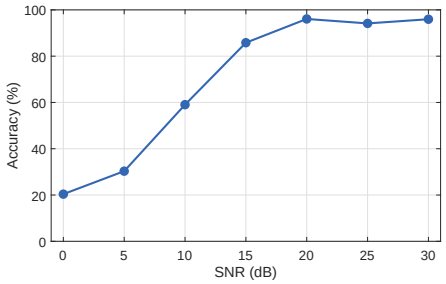


Fig. S6: Identification accuracy under different SNRs (WiFi, 20-shot, MS).

IV. Quantitative Feature-Space Analysis

Corresponding to the feature visualization (now reported in the main text Fig. 3), we report the Silhouette coefficient (SC), the normalized mutual information (NMI), and the adjusted mutual information (AMI) of the learned features under the 20-shot setting in Table S2. The quantitative metrics are consistent with the visualization: on both datasets the features form well-separated and compact clusters, with WiFi exhibiting the higher values, which is consistent with the larger inter-class separation visible in Fig. 3.

TABLE S2: Quantitative feature-space analysis under the 20-shot setting.

Metrics	LoRa	WiFi
SC ($\uparrow$)	0.5605	0.6261
NMI ($\uparrow$)	0.9382	0.9608
AMI ($\uparrow$)	0.8934	0.9419

SC: Silhouette coefficient; NMI: normalized mutual information; AMI: adjusted mutual information.

V. Computational Complexity

We report the parameter count, training time, inference latency, and peak training GPU memory of the proposed method and AMAE in Table S3. The proposed method uses 4.24 M parameters, i.e., about 61% fewer than AMAE (10.93 M), at the cost of a longer training time (13.29 vs. 8.48 s/epoch), a slightly higher inference latency (0.41 vs. 0.33 ms/sample, about 27% higher), and a larger peak training GPU memory (10.35 vs. 4.90 GB), which mainly stems from the multi-layer aggregation and the attention-based decoder. In absolute terms the overhead is modest and is paid once at training time; the inference latency in particular stays below one millisecond per sample. During inference, only the encoder and the classifier are retained.

TABLE S3: Computational overhead comparison.

Method	Params (M)	Training time (s/epoch)	Inference latency (ms/sample)	Peak GPU memory (MB)
Proposed	4.24	13.29	0.41	10349.66
AMAE	10.93	8.48	0.33	4898.24

VI. Paired Wilcoxon Signed-Rank Test Details

Since all methods are evaluated under the same 50 random seeds, the paired Wilcoxon signed-rank test is conducted on the improvements over the best-performing comparative method in each setting. The detailed results are listed in Table S4. The test is two-sided, paired over the same 50 seeds, and the p-values are Holm-corrected across the nine settings. The improvements are statistically significant ($p \leq 2.0\times10^{-3}$) in 8 of the 9 settings; the only exception is the ADS-B 10-shot setting, where the improvement over AMAE is not significant ($p = 0.25$), i.e., the case with the smallest margin and the largest relative variance, so no advantage is claimed there.

TABLE S4: Paired Wilcoxon signed-rank test results (50 shared seeds).

Setting	Best-performing baseline	Improvement	p (Holm)
ADS-B 10-shot	AMAE 66.05±3.54%	+1.07%	0.25
ADS-B 20-shot	AMAE 82.27±2.22%	+2.43%	2.0×10^{-3}
ADS-B 30-shot	AMAE 86.90±1.98%	+2.48%	5.0×10^{-5}
LoRa 10-shot	CSEE 80.62±4.76%	+4.00%	8.6×10^{-4}
LoRa 20-shot	DIPA 89.25±1.57%	+3.14%	3.0×10^{-4}
LoRa 30-shot	DIPA 92.43±1.76%	+2.65%	8.6×10^{-4}
WiFi 10-shot	DIPA 85.61±1.64%	+4.81%	1.5×10^{-7}
WiFi 20-shot	DIPA 91.47±0.95%	+5.60%	7.2×10^{-9}
WiFi 30-shot	DIPA 96.14±0.44%	+2.29%	9.2×10^{-9}

VII. Gradient-Handling Ablation

We compare the stop-gradient design (adopted in the manuscript) with the EMA-teacher variant and the variant without stop-gradient on the full-view branch. The results are listed in Table S5. The adopted stop-gradient design performs on par with the "Stop-gradient + EMA teacher" variant, while removing the stop-gradient clearly degrades the identification accuracy (84.70±2.68% → 69.47±5.53% on ADS-B and 97.07±1.47% → 85.52±3.98% on WiFi), confirming the necessity of the stop-gradient operation.

TABLE S5: Gradient handling of the full-view branch.

Variant	ADS-B (SS)	WiFi (MS)
Stop-gradient (default)	84.70±2.68%	97.07±1.47%
Stop-gradient + EMA	85.42±2.51%	97.48±1.22%
w/o stop-gradient	69.47±5.53%	85.52±3.98%

VIII. Projection-Module Ablation (MLP vs. Linear)

We compare the learnable linear projection ϕ_l with a two-layer MLP projection in the hierarchical feature aggregation. The results show that the performance difference is negligible (84.35±2.75% vs. 84.70±2.68% under the ADS-B 20-shot setting), indicating that the gain mainly stems from the aggregation mechanism rather than the projection capacity.